Icarus 299 (2018) 133–147

Contents lists available at ScienceDirect

Icarus

journal homepage: www.elsevier.com/locate/icarus

Two-phase convection in Ganymede's high-pressure ice layer — Implications for its geological evolution

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ARTICLE INFO

Article history:
Received 27 March 2017
Revised 14 July 2017
Accepted 21 July 2017
Available online 25 July 2017

Keywords:
Ganymede
Satellites, dynamics
Thermal histories
Interiors
Ices

ABSTRACT

Ganymede, the largest moon in the solar system, has a fully differentiated interior with a layer of high-pressure (HP) ice between its deep ocean and silicate mantle. In this paper, we study the dynamics of this layer using a numerical model of two-phase ice–water mixture in two-dimensional Cartesian geometry. While focusing on the generation of water at the silicate/HP ice interface and its upward migration towards the ocean, we investigate the effect of bottom heat flux, the layer thickness, and the HP ice viscosity and permeability. Our results suggest that melt can be generated at the silicate/HP ice interface for small layer thickness ($\lesssim 200$ km) and high values of heat flux ($\gtrsim 20$ mW m^{-2}) and viscosity ($\gtrsim 10^{15}$ Pa s). Once generated, the water is transported through the layer by the upwelling plumes. Depending on the vigor of convection, it stays liquid or it may freeze before melting again as the plume reaches the temperate (partially molten) layer at the boundary with the ocean. The thickness of this layer as well as the amount of melt that is extracted from it is controlled by the permeability of the HP ice. This process constitutes a means of transporting volatiles and salts that might have dissolved into the melt present at the silicate/HP ice interface. As the moon cools down, the HP ice layer becomes less permeable because the heat flux from the silicates decreases and the HP ice layer thickens.

1. Introduction

The exploration of ocean worlds is prompted by the question of the existence of extraterrestrial life and its emergence in places where water is or has been present. A lot of attention is presently given to places such as Europa and Enceladus for which the interior structure models predict a direct contact of the ocean (buried below the icy shell) with the silicate mantle (Schubert et al., 2009; Iess et al., 2014; Thomas et al., 2016; Čadek et al., 2016). Larger icy moons, such as Ganymede, Titan, or Callisto, are considered to host sufficient amounts of water that a layer of high-pressure (HP) ice exists between the deep ocean and the silicate mantle (e.g. Consolmagno and Lewis, 1976; Sohl et al., 2002; Sotin and Tobie, 2004; Hussmann et al., 2015) that prevents a direct contact of ocean water with silicates.

This study focuses on Ganymede, the largest icy moon in the solar system. Its moment of inertia (MoI) factor of 0.310 (Anderson et al., 1996) is consistent with a fully differentiated interior that comprises a thick H_2O layer, a silicate mantle, and a liquid iron core where Ganymede's intrinsic magnetic field is generated (Kivelson et al., 1996; Schubert et al., 1996). In addition to the internal field, an induced magnetic signal was inferred by the Galileo spacecraft (Kivelson et al., 2002) that can be explained by the presence of an internal, electrically conductive salty ocean. Recently, the HST observations by Saur et al. (2015) revealed weak auroral ovals oscillation in agreement with the existence of a deep ocean. Based on the phase diagram of water ice, this liquid water ocean is sandwiched between the outer ice I shell (less dense than water) and the deep high-pressure layer composed of ice VI, and possibly overlaid by ice V and ice III. The thickness of the HP ice layer is correlated with the thickness of the ice I shell (Vance et al., 2014): a thinner shell results in a warmer and thicker ocean and thus a thinner HP ice layer. Unfortunately, the ice shell thickness is not yet known but it may be estimated by the JUICE mission (Grasset et al., 2013). This would allow us to choose among the interior models that are described in the present study.

While the bulk of the deep HP ice layer prevents a direct contact between the ocean and the silicate mantle, the heat and ma-

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http://dx.doi.org/10.1016/j.icarus.2017.07.018
0019-1035/

terial exchange between these two layers might still be possible, but the exchange processes are poorly understood. Even though laboratory experiments on the viscosity of HP ices suggested that convection processes were possible (Poirier et al., 1981), numerical studies are rare as the expected presence of melt in the cold thermal boundary layer below the ocean/HP ice interface prevents the use of standard solid-state convection models. Indeed, recent three-dimensional numerical simulations of thermal convection in the HP ice layer indicate the occurrence of melting for a broad range of model parameters (Choblet et al., 2017). However, melt transport is not included in these models and instantaneous melt extraction is hypothesized.

In the present paper, we study the dynamics of the high-pressure ice layer of Ganymede using two-dimensional numerical simulations of two-phase convection that allow us to address the generation of meltwater and its transport through the convecting solid ice that was neglected in the previous study. We examine the conditions that lead to melting and investigate the fate of generated meltwater. We aim to determine the ability of Ganymede's HP ice layer to transfer not only the heat produced by the decay of radioactive elements contained in the silicate mantle but also – through melting processes – some material from the silicates to the deep ocean. Better understanding of transport processes through Ganymede's HP ice layer will shed light on the differentiation processes that have operated on Ganymede. For example, salts could be leached by water interacting with the silicates and salt transport through the HP ice layer within the upwelling plumes could significantly contribute to the salinity of the deep ocean. This might explain its conductivity, an important parameter that controls the amplitude of the induced magnetic field. The described process may also provide a way to transport more volatile species, the result of outgassing of the silicate core. The structure of this paper is as follows: We provide an overview of the assumed physical model and the chosen numerical method in Section 2. The results of numerical simulations are then described and synthesized in Section 3. We discuss the limitations of our model and compare our findings with the previously published results in Section 4. Finally, the implications of our results for the dynamics of Ganymede's HP ice layer are summarized in Section 5.

2. Model

2.1. Governing equations

To model the generation and transport of liquid water in a convecting HP ice layer, we consider the layer to behave as a mixture of two components – solid ice matrix (subscript i) and liquid water (subscript w). We further assume that the mixture can exist in two distinct states: **cold ice** (temperature is below the melting point, no liquid water is present) or **temperate** (partially molten) **ice** (temperature equals the melting point and interstitial liquid water can be present). The dynamics of such a mixture can be described by the equations derived in Souček et al. (2014). Depending on the connectivity of the interstitial water veins system, meltwater can either percolate through the convecting matrix (if the pores are interconnected) or be locked within the deforming ice and advected by it (if the pores are not interconnected). In the temperate parts of Earth's glaciers, meltwater percolates through the partially molten porous ice by a system of interconnected veins (e.g. Gusmeroli et al., 2010). In the HP ice layers, we expect water percolation to prevail within the hot thermal plumes.

For the water percolation through a solid ice matrix, we consider the two-phase equations in the zero compaction length approximation (e.g. Scott and Stevenson, 1989; Spiegelman, 1993). In this approximation, we assume that the ice deformation does not affect the porous fluid flow, which is then driven solely by the buoyancy due to density difference between the two phases. The effect of this approximation on the water extraction from a temperate ice I layer was investigated by Souček et al. (2014) in a one-dimensional setting. They have found that the neglect of the coupling terms leads to a change in the form of porosity propagation, however, the timescales for water extraction were of the same order in both cases. Therefore we find it adequate to use the reduced model for this study. In this setting, we solve the following equations:

$$\nabla \cdot (\phi \mathbf{v}_\mathrm{w} + (1-\phi)\mathbf{v}_\mathrm{i}) = \frac{\Delta\rho}{\rho_\mathrm{w}\rho_\mathrm{i}} r \ , \tag{1a}$$

$$\nabla\Pi = -\phi\Delta\rho\mathbf{g} - (1-\phi)\rho_\mathrm{i}\alpha\Delta T\mathbf{g} + \nabla\left(\frac{(1-\phi)\mu_\mathrm{i}}{\phi}(\nabla\cdot\mathbf{v}_\mathrm{i})\right) + \nabla\cdot\left((1-\phi)\mu_\mathrm{i}\left[\nabla\mathbf{v}_\mathrm{i} + (\nabla\mathbf{v}_\mathrm{i})^\mathrm{T} - \frac{2}{3}(\nabla\cdot\mathbf{v}_\mathrm{i})\mathbf{I}\right]\right) \ , \tag{1b}$$

$$\mathbf{v}_\mathrm{w} = \mathbf{v}_\mathrm{i} - \frac{K(\phi)}{\mu_\mathrm{w}\phi}(1-\phi)\Delta\rho\mathbf{g} \ , \tag{1c}$$

$$\frac{\partial\phi}{\partial t} + \nabla\cdot(\phi\mathbf{v}_\mathrm{w}) = \frac{r}{\rho_\mathrm{w}} \ , \tag{1d}$$

$$\rho_\mathrm{i} c_\mathrm{p}\frac{\partial T}{\partial t} + \rho_\mathrm{i} c_\mathrm{p}\mathbf{v}_\mathrm{i}\cdot\nabla T + Lr = k\nabla^2 T \ . \tag{1e}$$

Here ϕ is the porosity (volume fraction of water in the mixture), $\mathbf{v}_\mathrm{w}$ is the percolating water velocity, $\mathbf{v}_\mathrm{i}$ is the convecting ice velocity, ρ_i is the ice density, ρ_w is the water density, $\Delta\rho=\rho_\mathrm{i}-\rho_\mathrm{w}$ is the density difference between the two phases, which is assumed constant throughout the HP ice layer and taken equal to the average between the bottom and top interface values (cf. Table 2) – we discuss the effect of this assumption in Section 4.2; r is the melt production rate (positive for melting), $\Pi=P_\mathrm{w}-P_\mathrm{i}^\mathrm{ref}$ is the excess water pressure with respect to the hydrostatic equilibrium pressure of pure ice ($P_\mathrm{i}^\mathrm{ref}$), $\mathbf{g}$ is the gravity, α is the ice thermal expansivity, $\Delta T=T-T_\mathrm{m}$ is the temperature (T) difference from the melting temperature (T_m), μ_i is the ice shear viscosity, $K(\phi)$ is the porosity-dependent permeability, μ_w is the water shear viscosity, t is the time, c_p is the isobaric specific heat of ice, L is the latent heat of ice, and k is the ice thermal conductivity. The appropriate values of all parameters are discussed in Sections 2.2–2.3 and summarized in Tables 1 and 2.

Eq. (1a) expresses the mass balance of the whole mixture – note that even though the individual phases are considered incompressible (ρ_i=const., ρ_w=const.), the two-phase mixture as a whole can dilate/compact when melting/freezing occurs ($r\neq 0$). Eq. (1b) is the linear momentum balance of matrix – the first two terms on the right-hand side are the phase-change and thermal buoyancy, respectively, the third term corresponds to ice matrix compaction with an effective (porosity-dependent) bulk viscosity $\zeta\sim\frac{\mu_\mathrm{i}}{\phi}$ (cf. Ricard et al., 2001), and the last term corresponds to standard viscous shear deformation. Water velocity is computed from a simplified Darcy's law (Eq. 1c), where the simplification lies in the neglect of driving pressures other than those due to density difference. Water percolation through the convecting ice matrix is thus driven solely by the density difference between the two phases. Eq. (1d) is the mass balance of fluid in the form of an advection–reaction equation for porosity. Finally, Eq. (1e) is the internal energy balance of the mixture as a whole. Depending on whether the ice at a given point is cold ($T < T_\mathrm{m}(P)$, $r = 0$, ϕ=0) or temperate ($T = T_\mathrm{m}(P)$, $\phi \geq 0$), this equation either governs the evolution of temperature, or provides the melting/freezing rate r, respectively. Note that we use a zero porosity limit of the full two-phase energy balance (cf. Souček et al., 2014) by neglecting the advection of heat by liquid water and we do not consider adiabatic

Table 1
List of parameters used in the present study. $\bullet^{s}$ denotes the value at the silicate/HP ice interface, $\bullet^{*}$ denotes the investigated model parameters (Sections 3.2–3.5). If not specified otherwise, the values of material parameters correspond to ice VI.

Symbol	Physical property	Value	Unit
P^{s}	Pressure	1.6	GPa
T_{m}^{s}	Melting temperature[a]	332	K
g	Gravity acceleration	1.6	$m\,s^{-2}$
α	Thermal expansion[b]	1.46×10^{-4}	K^{-1}
c_{p}	Specific heat[c]	2850	$J\,kg^{-1}\,K^{-1}$
ρ_{i}^{s}	Ice density at (P^{s}, T_{m}^{s})[b]	1390	$kg\,m^{-3}$
ρ_{w}^{s}	Water density at (P^{s}, T_{m}^{s})[a,b]	1305	$kg\,m^{-3}$
k	Thermal conductivity[d]	1.58	$W\,m^{-1}\,K^{-1}$
L	Latent heat[a]	360	$kJ\,kg^{-1}$
μ_{w}	Water shear viscosity[e]	2×10^{-3}	Pa s
K_{0}	Reference permeability[f]	10^{-9}	m^{2}
n	Permeability exponent[f]	2	–
Q	Activation energy of ice viscosity[g,h]	30	$kJ\,mol^{-1}$
μ_{0}^{*}	Reference ice shear viscosity[g,h]	10^{14}–10^{17}	Pa s
H^{*}	HP ice layer thickness	100–400	km
q_{s}^{*}	Heat flux from silicates	10–40	$mW\,m^{-2}$
ϕ_{c}^{*}	Percolation threshold	1–5	%

[a] Bridgman (1912; 1937)
[b] Bezacier et al. (2014)
[c] Tchijov (2004)
[d] Ross et al. (1978)
[e] Lide (2004)
[f] Golden et al. (2007)
[g] Sotin et al. (1985)
[h] Durham and Stern (2001)

decompression. We discuss the effect of these approximations in Section 4.2.

The system of governing equations (1) is supplemented with the following boundary conditions. Free slip condition is prescribed on top and side boundaries in the form:

$$\mathbf{v}_{i}\cdot\mathbf{n}=0 \tag{2a}$$

$$\left[(1-\phi)\mu_{i}\left(\frac{1}{\phi}(\nabla\cdot\mathbf{v}_{i})\mathbf{I}+\nabla\mathbf{v}_{i}+(\nabla\mathbf{v}_{i})^{T}-\frac{2}{3}(\nabla\cdot\mathbf{v}_{i})\mathbf{I}\right)\cdot\mathbf{n}\right]_{t}=\mathbf{0}\,, \tag{2b}$$

where $\mathbf{n}$ is the outer normal to the surface and $\bullet_{t}$ denotes the tangential component of a vector. This condition is a modification of the standard free slip condition with a new term that can be interpreted as a volume deformation with a porosity dependent bulk viscosity (cf. above), and it is also a natural boundary condition of Eq. (1b) in the finite element method as it eliminates the boundary integrals in the weak form. No slip condition is prescribed for the ice velocity on the bottom boundary:

$$\mathbf{v}_{i}=\mathbf{0}\,. \tag{3}$$

The excess water pressure Π is determined up to a constant, fixed by setting $\Pi=0$ in one point of the computational domain. No boundary condition is required for porosity ϕ, since Eq. (1d) is hyperbolic and no inflow boundary is present in the current setting (e.g. Evans, 1998). Free outflow of water from the domain is allowed at the top boundary (interface with the ocean) while the bottom and side boundaries are kept impermeable for water:

$$\mathbf{v}_{w}\cdot\mathbf{n}=0\,. \tag{4}$$

Finally, we fix temperature at the top boundary:

$$T=T_{m}^{o}\,, \tag{5}$$

which is the melting temperature for a given pressure (Table 2) and we prescribe a fixed heat flux from the silicate mantle at the bottom boundary:

$$k\nabla T=-\mathbf{q}_{s}\,, \tag{6}$$

while the side boundaries are held insulating:

$$\nabla T\cdot\mathbf{n}=0\,. \tag{7}$$

Depending on the overall thermal state of the H_2O layers, the initial temperature T_0 could be a constant equal to the melting temperature at the interface with the ocean (T_{m}^{o}) or a pressure-dependent melting curve ($T_{m}(P)$). The former case corresponds to a very efficient cooling due to water (and heat) extraction to the ocean, while the latter corresponds to inefficient cooling by solid state convection. Having no constraints, we have tested both initial conditions and found the same steady-state for both (Section 3.6). In the majority of simulations, however, the initial temperature profile is set to the top boundary melting temperature, $T_0=T_{m}^{o}$.

2.2. Model parameters specific to Ganymede

In the present study, we use the approach of Vance et al. (2014) that employs the equation of state (EoS) of water with different amount of sulfates to compute the radius of internal interfaces. Ganymede is assumed to be fully differentiated and all resulting interior structure models are consistent with the value of the MoI factor equal to 0.310 (Anderson et al., 1996). The nominal model has an iron-rich liquid core at the Fe/FeS eutectic (density of 7030 $kg\,m^{-3}$) and an anhydrous silicate mantle with a density of 3250 $kg\,m^{-3}$. We use the EoS of the different H_2O phases to compute the density profile from the surface to the silicate/HP ice interface. The inversion of mass and MoI factor provides the radius of the core and of the silicate/HP ice interface. Changing the assumed density values may slightly modify the latter. Vance et al. (2014) investigated two parameters: the thickness of the ice I layer and the amount of $MgSO_4$ in the ocean. The radius of the silicate mantle varies between 1814 and 1836 km when only the ice I thickness is varied and between 1738 and 1836 km when the two parameters are varied. However, this change would not modify the main conclusions of this study on the dynamics and structure of the HP ice layer. Also, since only a few constraints exist, we decided not to investigate a large range of parameters but rather to focus on the dynamics of the HP ice layer. At the interface between the hydrosphere and the silicate mantle (with radius of 1830 km), the pressure is equal to 1.6 GPa and the **gravity acceleration** is on the order of 1.7 $m\,s^{-2}$, compared to 1.4 $m\,s^{-2}$ at the surface. For the sake of simplicity, a constant value of g=1.6 $m\,s^{-2}$ is considered in the

Table 2
Pressure, melting temperature, ice and water densities at the ocean/HP ice interface (denoted by $\bullet^{o}$) evaluated for different HP ice layer thicknesses H. The average density values in the particular HP ice layer (denoted by $\bar{\bullet}$) are computed from the values at both interfaces (marked by $\bullet^{s}$ and $\bullet^{o}$). The values used in numerical simulations are marked in bold.

H [km]	P^{o} [GPa]	T_{m}^{o} [K]	ρ_{i}^{o} [$kg\,m^{-3}$]	ρ_{w}^{o} [$kg\,m^{-3}$]	$\bar{\rho}_{i}$ [$kg\,m^{-3}$]	$\bar{\rho}_{w}$ [$kg\,m^{-3}$]	$\Delta\bar{\rho}$ [$kg\,m^{-3}$]
100	1.378	**321**	1376	1279	**1383**	**1292**	**91**
200	1.155	**309**	1363	1254	**1377**	**1280**	**97**
300	0.933	**295**	1350	1226	**1370**	**1266**	**105**
400	0.710	**281**	1335	1205	**1363**	**1255**	**108**

HP ice layer. Several values of the HP ice layer **thickness** H are investigated, from 100 km to 400 km (Table 2).

The **heat flux** q_s at the silicate/HP ice interface varies through Ganymede's evolution and depends on the amount of radioactive elements and the means of heat transfer (convection/conduction) in the silicate mantle (Tobie et al., 2006). We consider a plausible range of heat flux values from 10 to 40 mW m^{-2}. The low value is characteristic for the period just after accretion and differentiation when radioactive heating has not yet diffused to the interface, but also for the present time when the amount of radioactive elements (in particular ^{40}K) has decayed significantly. The largest value is expected when convection initiated in the silicate mantle after an initial stage of progressive warming by thermal diffusion of radioactive heat (Kirk and Stevenson, 1987; Tobie et al., 2006).

2.3. Model parameters specific to ice VI and liquid water

The parameters specific to ice VI that are present in Eq. (1) include melting temperature, thermal expansion, specific heat, density, thermal conductivity, latent heat, permeability, and viscosity. Moreover, values for liquid water density and viscosity are required.

The **melting temperature** of ice VI (Bridgman, 1912; 1937) is computed at the silicate/HP ice interface (T_m^s, Table 1) and at the ocean/HP ice interface (T_m^o, Table 2). For the sake of simplicity, we use a linear interpolation between these two values for the HP ice layer interior. It is worth noting that the diffusive heat flux associated with the thermal gradient along the melting curve is about two orders of magnitude lower than the heat flux coming from the silicate mantle.

The value of the **thermal expansion coefficient** α of ice VI at temperature close to the melting point was recently determined by Bezacier et al. (2014). Their value of 1.46×10^{-4} K^{-1}, which is used in this study, is about 40% larger than values used previously (e.g. Grindrod et al., 2008).

The **specific heat** c_p of ice VI has been calculated by Tchijov (2004) for different polymorphs of ice by using the knowledge of phase transitions and by anchoring the values to the experimental data on ice VII (Fei et al., 1993) and ice I. The value for ice VI at its melting point is c_p=2850 J kg^{-1} K^{-1}, which is significantly higher than the value of 1925 J kg^{-1} K^{-1} used by Grasset and Sotin (1996).

The **density of ice VI** is calculated by using the EoS determined by Bezacier et al. (2014). The **density of water at the melting point** is calculated by using the ice VI EoS and values of the volume change between the ice and water measured by Bridgman (1912; 1937).

Thermal conductivity of ice VI was measured by Ross et al. (1978) for temperatures between 100 and 300 K and under pressures of up to 2.2 GPa. Extrapolation of their values along the melting curve – between (300 K, 1 GPa) and (331 K, 1.6 GPa) – gives values of 1.54 W m^{-1} K^{-1} and 1.62 W m^{-1} K^{-1}. These variations are lower than the uncertainties reported in Andersson and Inaba (2005). In the present study, we use an intermediate value of k=1.58 W m^{-1} K^{-1}. Note that previous studies (e.g. Grasset and Sotin, 1996) have used a much lower value of 0.56 W m^{-1} K^{-1} inferred from values given in Hobbs (1974).

The **latent heat** can be inferred from the slope of the melting curve using the Clausius–Clapeyron equation. Bridgman (1912; 1937) provides values of the latent heat using this equation with the volume change determined in his high-pressure experiments. Even though the latent heat varies along the melting curve, for the sake of simplicity, the present study uses the value of 360 kJ kg^{-1} computed at P^s=1.6 GPa.

Permeability, which is the ability of a material to transmit fluids, plays a key role in determining the veins system connectivity and therefore the efficiency of water extraction. The ice permeability generally depends on numerous parameters, such as the geometric properties of the pore system, pressure in the liquid, temperature, composition, deformation history etc. (cf. Lliboutry, 1996). For terrestrial sea ice, an abrupt permeability decrease for porosities below the 'percolation threshold' $\phi_c \sim 5\%$ was reported by Golden et al. (1998) which corresponds to the closure of the system of veins in the ice and the subsequent drop in its connectivity. In the absence of better knowledge, we assume that the HP ice in the planetary layers behaves in a similar manner even though the behavior of partial melt might be different due to the strong difference in pressure. We thus use the following equation:

$$K(\phi) = \begin{cases} K_0(\phi-\phi_c)^n & \phi \geq \phi_c \\ 0 & \phi < \phi_c \end{cases}, \tag{8}$$

where ϕ_c is the percolation threshold (here considered to be between 1% and 5%) and K_0, n are experimentally determined constants for which we use the values inferred for terrestrial ice.

The **ice viscosity** is a critical parameter that controls the dynamics of the HP ice layer. It has been measured by two groups (Sotin et al., 1985; Durham et al., 1996) and we provide a detailed comparison of their findings in Appendix A. Both groups report that the isoviscous curves are almost parallel to the melting curve. This observation justifies the use of the following expression to describe the temperature and pressure dependence of ice viscosity:

$$\mu_i = \mu_0 \exp\left[\frac{Q}{R}\left(\frac{1}{T}-\frac{1}{T_m(P)}\right)\right]. \tag{9}$$

Here μ_0 is the reference viscosity at the melting temperature, Q is the activation energy, and R is the universal gas constant. Extrapolation of values obtained in the high shear stress laboratory experiments to the low shear stresses appropriate for convection and the need to take into account the potential changes in the major creep mechanisms lead us to investigate a large range of melting point viscosity values from 10^{14} to 10^{17} Pa s, with the reference value of 10^{15} Pa s. Moreover, the presence of partial melt softens the ice – we investigate the possible effect of this softening by modifying the ice viscosity given by Eq. (9) as follows (cf. Tobie et al., 2003):

$$\mu_i^\phi = \begin{cases} \mu_i \exp(-\gamma_m\phi) & \phi \leq \phi_\gamma \\ \mu_i \exp(-\gamma_m\phi_\gamma) & \phi > \phi_\gamma \end{cases}, \tag{10}$$

with ϕ_γ=5% and γ_m=45. These values correspond to one order of magnitude viscosity reduction per five percents of melt reported by De La Chapelle et al. (1999) for ice I. For simplicity and lack of better knowledge, we assume the same behavior for ice VI.

2.4. Numerical method

We consider a Cartesian computational domain of aspect ratio 2 which is kept constant throughout the simulations (in agreement with the boundary conditions for ice matrix, Eqs. 2 and 3). However, this assumption is not fully consistent with the free water outflow and the mass balance of the whole mixture (Eq. 1a). Neglecting the layer shrinking due to water outflow is equivalent to assuming that the outflowing water freezes. The average position of the ocean/HP ice interface is given by the intersection of the adiabatic profile in the ocean and the melting curve of HP ice (e.g. Grasset and Sotin, 1996). In equilibrium, any change of the position of this interface (associated with water outflow) would result in a temperature difference between the adiabatic profile and the melting curve which would lead to crystallization. Keeping the computational domain constant thus corresponds to the equilibrium assumption. Throughout this study, we use a structured mesh that consists of 200 × 100 crossed diagonal triangular elements and has a refined layer of height of 0.05 × H at the bottom boundary. This mesh contains 91,000 triangular cells and for the reference layer

Table 3
List of simulations with the reference case marked in bold. Rayleigh number for basally-heated convection is defined as $\mathrm{Ra}=\frac{\alpha \rho_i^2 c_p g q_s H^4}{k^2 \mu_0}$. The thickness of the temperate layer H^{temp} is horizontally averaged.

run #	μ_0 [Pa s]	γ_m –	H [km]	q_s [W/m^2]	ϕ_c [%]	T_0 –	Ra –	H^{temp} [km]
1	$\mathbf{10^{15}}$	**0**	**200**	**0.02**	**1**	$\mathbf{T_m^o}$	$\mathbf{1.6 \times 10^{10}}$	**15.7**
2	10^{14}	0	200	0.02	1	T_m^o	1.6×10^{11}	11.6
3	10^{16}	0	200	0.02	1	T_m^o	1.6×10^{9}	17.8
4	10^{17}	0	200	0.02	1	T_m^o	1.6×10^{8}	62.0
5	10^{15}	45	200	0.02	1	T_m^o	1.6×10^{10}	15.5
6	10^{15}	0	100	0.01	1	T_m^o	5.1×10^{8}	12.0
7	10^{15}	0	100	0.02	1	T_m^o	1.0×10^{9}	13.4
8	10^{15}	0	100	0.04	1	T_m^o	2.0×10^{9}	13.0
9	10^{15}	0	200	0.01	1	T_m^o	8.1×10^{9}	14.7
10	10^{15}	0	200	0.03	1	T_m^o	2.4×10^{10}	17.5
11	10^{15}	0	200	0.04	1	T_m^o	3.2×10^{10}	18.1
12	10^{15}	0	300	0.02	1	T_m^o	8.1×10^{10}	15.5
13	10^{15}	0	300	0.04	1	T_m^o	1.6×10^{11}	15.3
14	10^{15}	0	400	0.01	1	T_m^o	1.3×10^{11}	15.0
15	10^{15}	0	400	0.02	1	T_m^o	2.5×10^{11}	16.1
16	10^{15}	0	400	0.04	1	T_m^o	5.1×10^{11}	16.2
17	10^{15}	0	200	0.02	3	T_m^o	1.6×10^{10}	35.2
18	10^{15}	0	200	0.02	5	T_m^o	1.6×10^{10}	57.2
19	10^{15}	0	200	0.02	1	$T_m(P)$	1.6×10^{10}	16.5
20	10^{15}	0	400	0.02	1	$T_m(P)$	2.5×10^{11}	17.4

thickness of 200 km the element size is 2 km in the global mesh and 1 km in the refined bottom layer. We discuss the effect of resolution and aspect ratio in Section 4.3.

Numerical solution is obtained by using the open source finite element software package FEniCS (http://fenicsproject.org, Logg et al., 2012; Alnæs et al., 2015). We use discontinuous Galerkin method (Di Pietro et al., 2006) to discretize the porosity transport equation (1d), the Taylor–Hood elements (i.e. piece-wise linear functions for excess pressure and piece-wise quadratic functions for ice velocity, cf. Taylor and Hood, 1973) for the Stokes problem (Eqs. 1a–1b), and piece-wise quadratic functions for temperature (Eq. 1e) and fluid velocity (Eq. 1c). The Streamline Upwind/Petrov–Galerkin (SUPG) regularization (Brooks and Hughes, 1982) is applied for the advection part of the energy balance. For time discretization, we employ the implicit Euler method with an adaptive time-stepping controlled by the Courant–Friedrichs–Lewy (CFL) criterion. The assumed zero compaction length approximation allows us to decouple the two problems with different timescales (ice convection and water percolation) in a following manner – we solve the convection problem (Eqs. 1a, 1b, 1e) and for each convection time step, we solve the water percolation problem (Eqs. 1c, 1d) in a series of smaller time steps appropriate to the fluid velocity $\mathbf{v}_w$. The convection part of the numerical code was tested against the convection benchmark of Blankenbach et al. (1989) and successfully compared with the results of Vynnytska et al. (2013). The conservation of energy and mass is also verified at each time step. For numerical reasons (term $(1-\phi)/\phi$ in Eq. 1b), a small background value of porosity ($\phi = 0.5\%$) is permanently imposed, even for cold ice. This background value does not significantly affect the results since it is always smaller than the value of percolation threshold and therefore does not participate in the extraction process.

3. Results of numerical simulations

The complete list of performed simulations is given in Table 3 with the reference case marked in bold. Other (unvaried) parameters are summarized in Tables 1 and 2. We start with a detailed description of the reference case and then we proceed with the investigation of the effect of the main model parameters: melting point viscosity μ_0, HP ice layer thickness H, heat flux from the silicates q_s, percolation threshold ϕ_c, and the initial temperature profile T_0.

3.1. Reference simulation

We computed our reference solution with the layer thickness H=200 km, melting point viscosity μ_0=10^{15} Pa s, no porosity softening of viscosity (γ_m=0), bottom heat flux q_s=20 mW m^{-2}, percolation threshold ϕ_c=1%, and the initial temperature equal to the melting temperature at the interface with the ocean, T_0=T_m^o (simulation # 1 in Table 3). Fig. 1 shows the temperature profiles (maximum, minimum, and horizontal average; first column), difference between the melting temperature and the ice temperature (second column), porosity (third column), and the porosity profiles (maximum, minimum, and horizontal average; fourth column) at three different times (1, 3 and 10 Myr).

Panels in the top row depict the time shortly after the onset of convection ($t\sim$1 Myr) and show a few thermal plumes with no melt since temperature in the plume heads is about $\sim$1 K below the pressure melting point (cf. the red profile in the left panel). However, the bottom boundary is at the melting point and some patches of melt start to appear below the upwelling plumes. Panels in the next row ($t\sim$3 Myr) show a developed convective layer with a few ascending plumes. The average temperature of the layer is well below the melting point (cf. the grey profile in the left panel), however, temperature in some of the upwelling hot plumes reaches the melting point during the ascent which leads to melting in the plume heads, cf. red profiles in the left (temperature) and right (porosity) panels. The top of the layer is partially molten (temperate) with the average porosity $\phi_{av}\sim$1% and some water is extracted to the above lying ocean. The presence of a temperate layer inhibits the development of a cold thermal boundary layer. No cold downwelling can form and thus the convective pattern is characterized by hot upwelling plumes that originate in the hot thermal boundary layer at the interface with silicates and a passive global downwelling. At the bottom boundary, a small amount of melt is present.

We define steady-state as an equilibrium between the heat that comes from the underlying silicate mantle and the heat that is supplied to the overlying ocean by both conduction through the top temperate layer and ice melting and subsequent liquid water extraction. Because the conductive heat flow along the melting curve is approximately 2 orders of magnitude smaller than the incoming heat flux from the silicates, the majority of supplied heat is, in steady-state, consumed on ice melting. It is therefore to be understood as a steady-state of the HP ice layer (the average temperature in the layer is constant) but not as a steady-state of the whole moon. We can see that in such a steady-state, which is reached after $\sim$10 Myr (cf. the bottom row of Fig. 1), the layer can be divided into three distinct regions: (i) the bottom boundary (interface with the silicates) where T_{av}=T_m and a thin layer with $\phi_{av} \lesssim 1\%$ is present; (ii) the convective interior where $T_{av} < T_m$ but a small amount of partial melt is present in some of the upwelling plume heads; and (iii) the top boundary (interface with ocean) with a temperate layer of approximately 16 km (cf. Table 3) where T_{av}=T_m and $\phi_{av}\sim$1%. Even though a thin temperate layer is present at the bottom boundary and water is extracted to the overlying ocean, there is no direct fluid path through the whole layer.

Water transport through the layer and its extraction into the ocean

We now investigate the ascent of melt through the layer in more detail. Fig. 2 shows part of the computational domain ($x \in \langle 250, 350\rangle$ km) in succeeding time instants around $t\sim$10 Myr and traces the ascent of two plumes that originated at the bottom boundary. At the beginning of the ascent ($t\sim$9.86 Myr), there is no melt in the first plume, while there is small amount of melt

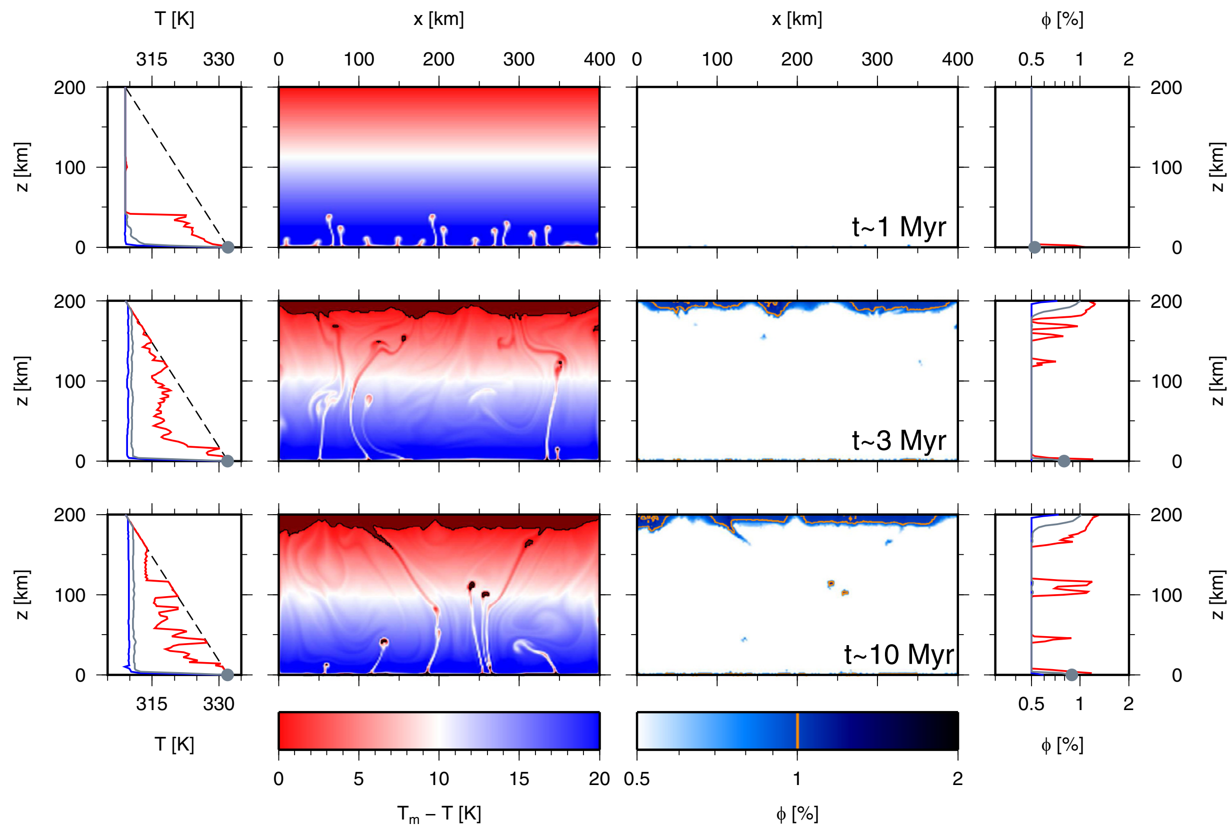

Fig. 1. Results for the reference simulation (# 1 in Table 3) at three different times. First column: temperature profiles – maximum (red), minimum (blue), horizontal average (grey), and melting temperature (dashed line); second column: difference between melting temperature and ice temperature; third column: porosity; fourth column: porosity profiles – maximum (red), minimum (blue), and horizontal average (grey). Dark red in temperature panels marks temperate ice ($T = T_m$), orange contour in porosity panels marks $\phi = 1\%$. The grey circles in the left and right panels mark the bottom boundary values of average temperature and porosity, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in the second plume. As the first plume rises into the upper half of the layer, some melt eventually appears in the plume head because melting temperature decreases with the increasing height (decreasing pressure). Due to combined buoyancy of temperature and melt, the second plume ascends faster than the first one and at $t \sim 9.95$ Myr, the two plumes merge and reach the temperate region below the top boundary with the ocean. Also, more melt is produced during this later phase of plume ascent.

Our results indicate that the amount of liquid in the plume head varies as the plume migrates upwards. One can write that the freezing rate of the liquid in the plume head, $-L\,(\mathrm{d}m/\mathrm{d}t)$ (where m is the mass of the liquid and L is the latent heat), is equal to the difference between the sensible heat as the upwelling brings the material closer to the melting point ($\mathrm{d}T_m/\mathrm{d}z$ is negative) and the cooling by the cold global downwelling:

$$-L\,\frac{\mathrm{d}m}{\mathrm{d}t} = S\,q + \pi\,R^2\,l\,\rho\,c_p\,w\,\frac{\mathrm{d}T_m}{\mathrm{d}z}\ . \tag{11}$$

Here S is the surface area of the plume head ($S = 2\pi\,R\,l$ in two dimensions with l a unit length perpendicular to the two-dimensional plane and R the radius of the plume), w is the vertical velocity, and q is the heat flux through the plume head which can be approximated by:

$$q = k\,\frac{T_m - T_{av}}{R}\ . \tag{12}$$

Just above the hot thermal boundary layer where the difference in temperature between the cold global downwelling material (T_{av}) and the melting temperature is largest, cooling is stronger than the warming by sensible heat and some liquid freezes. As the plume migrates upwards, warming by sensible heat prevails and melt is produced. This is well illustrated in Fig. 2 where the fate of plume is depicted in detail during its ascent from the bottom boundary layer to the top boundary which takes approximately 0.1 Myr.

Water extraction into the ocean occurs in outbursts of small amount of water when $\phi > \phi_c$ and is highly variable in time and space. Because the porosities are just slightly above the threshold value ($\phi - \phi_c \lesssim 0.5\%$), the corresponding extraction velocities are only on the order of a few meters per year.

3.2. Ice viscosity

Based on results from laboratory experiments (Sotin et al., 1985; Durham et al., 1996) which were extrapolated to low values of convective shear stresses (Appendix A), four different values of melting point viscosity μ_0 were considered: 10^{14}, 10^{15} (reference case), 10^{16}, and 10^{17} Pa s (simulations # 1–4 in Table 3). Fig. 3 shows the temperature and porosity and the corresponding profiles in a statistical steady-state for the first three simulations (10^{14}–10^{16} Pa s). Clearly, the reference viscosity plays a crucial role in the HP ice layer dynamics by significantly modifying the Rayleigh number. Decreasing the reference viscosity to $\mu_0 = 10^{14}$ Pa s leads to more vigorous convection and thus to more efficient heat transfer. As a result, the average temperature at the interface with silicates is below the melting point (cf. the grey circle in the top left panel of Fig. 3) and there is no meltwater generated at the bottom boundary. On the other hand, increasing the reference viscosity to $\mu_0 = 10^{16}$ Pa s results in establishment of two distinct convective cells with two upwellings stable in time. These upwellings are at the melting temperature (cf. the red curve in the bottom left

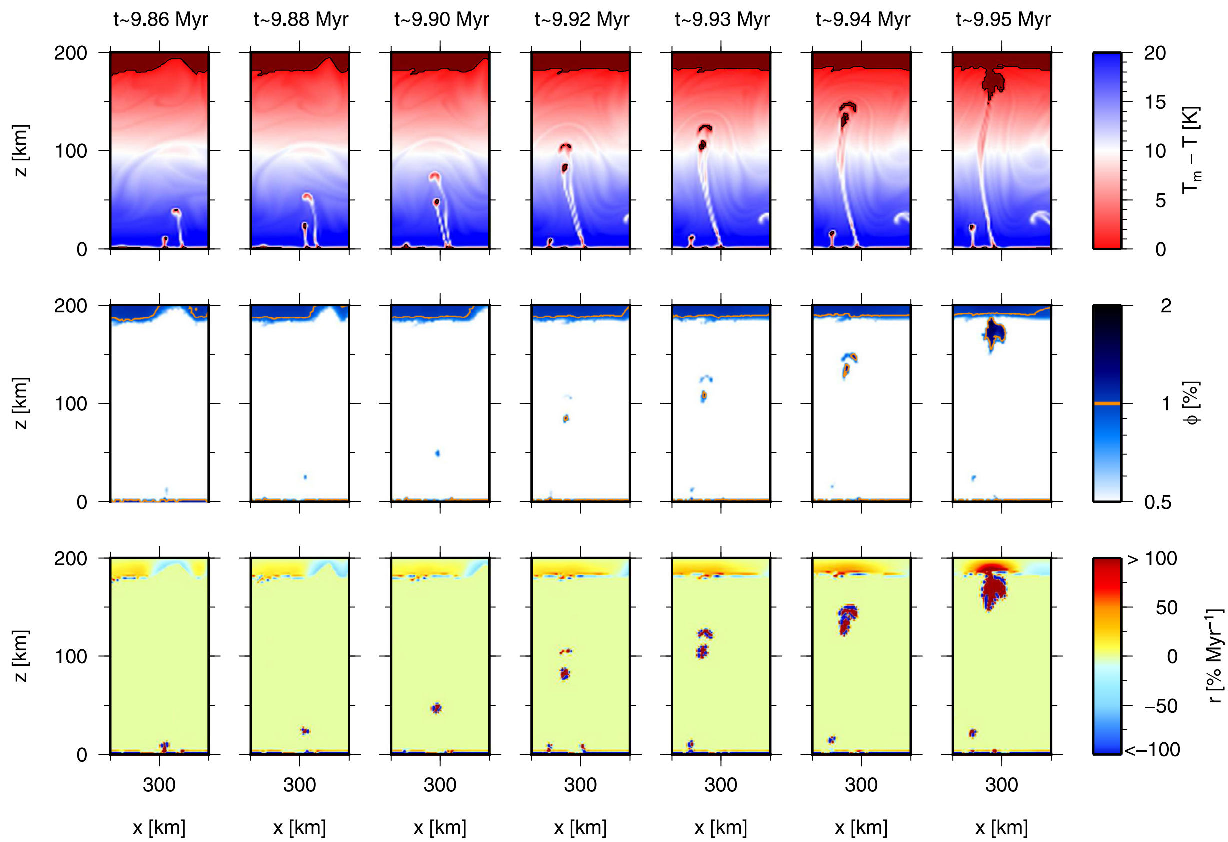

Fig. 2. Detail of the ascending plumes in the reference case. Difference between melting temperature and ice temperature (top), porosity (middle), melting/freezing rate (bottom).

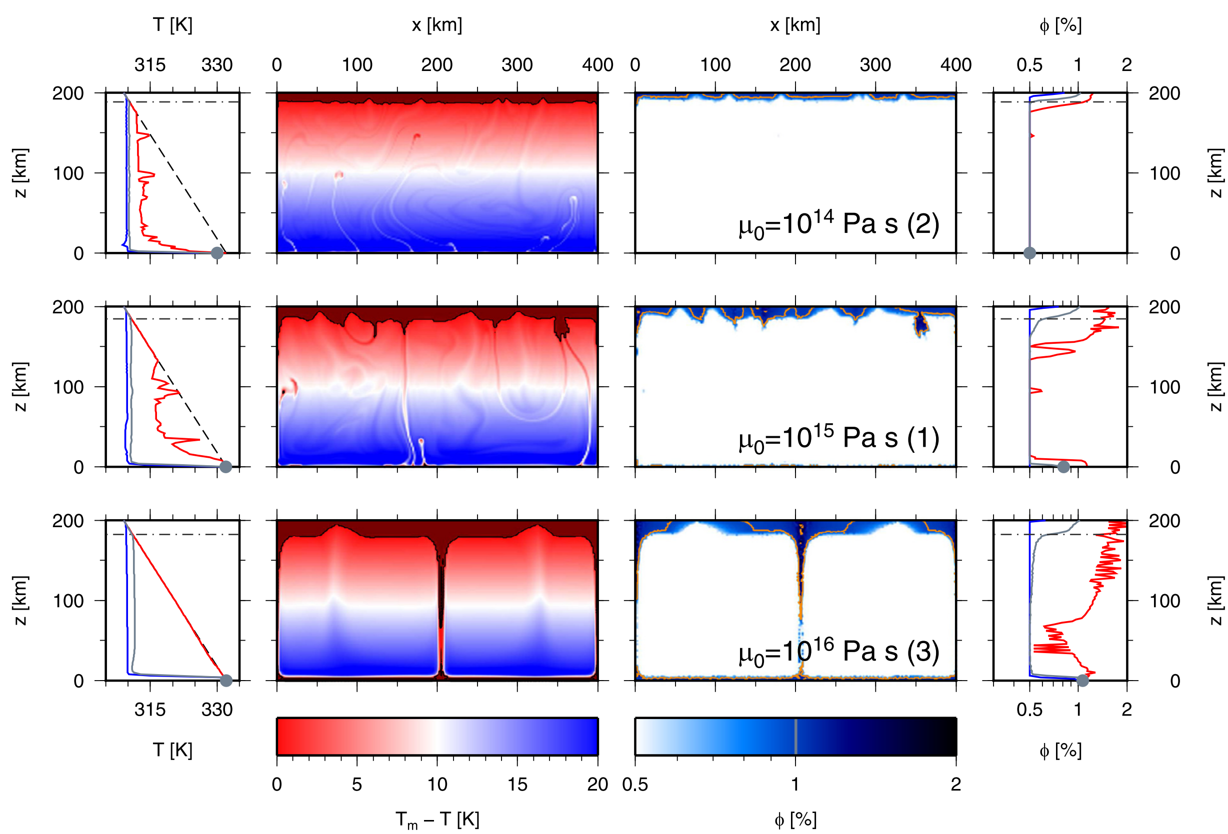

Fig. 3. The same as in Fig. 1 but for different melting point viscosities (10^{14}–10^{16} Pa s, top to bottom, numbers in parentheses refer to simulation numbers from Table 3) and in a statistical steady-state. Horizontal dot-dashed lines mark the extent of the top temperate layer (cf. Table 3).

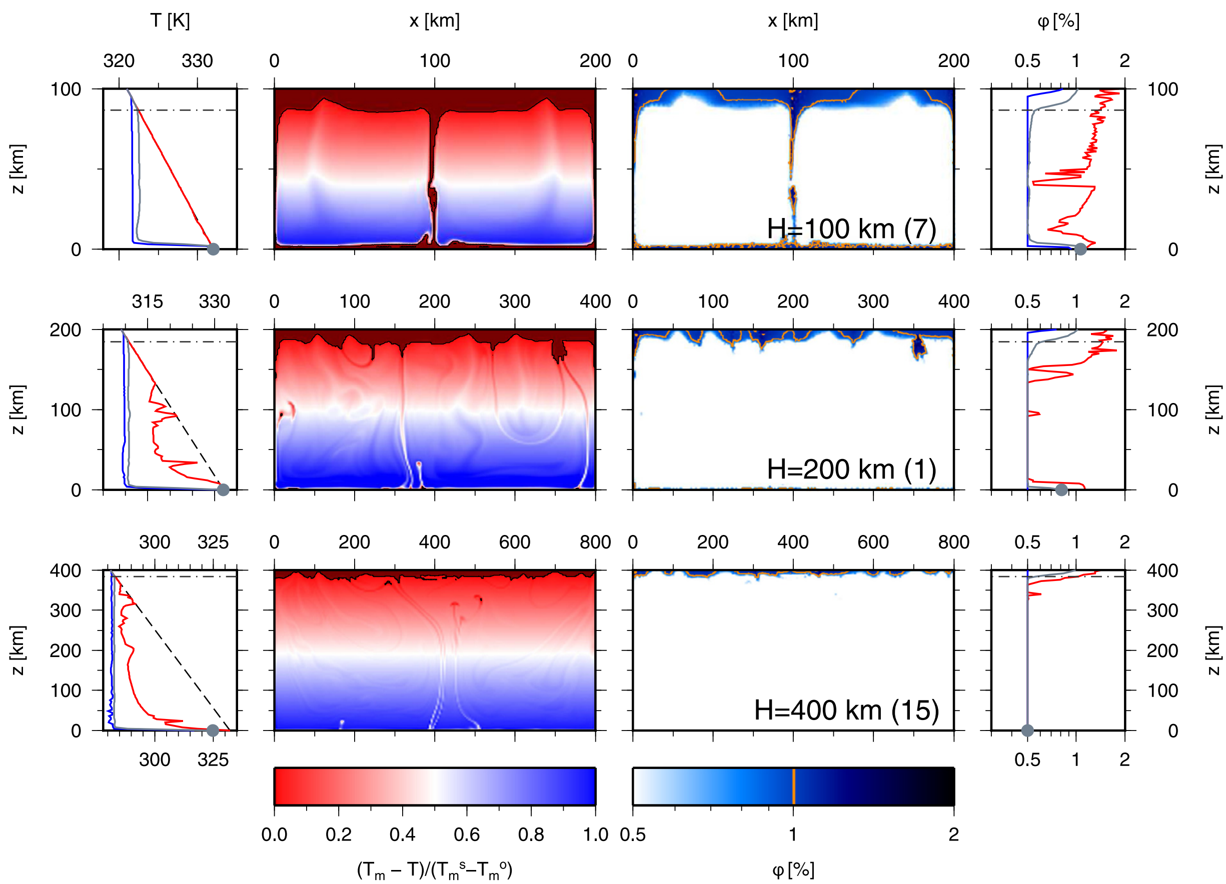

Fig. 4. The same as in Fig. 3 but for different layer thicknesses (100, 200 and 400 km, top to bottom, numbers in parentheses refer to simulation numbers from Table 3).

panel) and contain some amount of liquid water above the background value (cf. the red curve in the bottom right panel) thus forming a direct fluid path through the whole layer. Increasing the reference viscosity further to 10^{17} Pa s (not shown here) changes the convection pattern from two cells to one cell with more massive upwellings that ensure the meltwater transport through the layer. Let us note that one simulation was also run with the reference value of $\mu_0 = 10^{15}$ Pa s while taking into account the viscosity softening due to presence of melt (cf. Eq. 10) with $\gamma_m = 45$ (simulation # 5 in Table 3). However, the effect of this viscosity softening was negligible when compared with the effect of the reference viscosity μ_0 and thus we do not further consider it in this study.

3.3. Layer thickness

Four HP ice layer thicknesses were considered (100, 200, 300, and 400 km) together with the reference value of heat flux q_s = 20 mW m^{-2} (simulations # 1, 7, 12, and 15 in Table 3). Fig. 4 shows the temperature and porosity and the corresponding profiles in a statistical steady-state for three of these four simulations. We can see that changing the HP ice layer thickness changes the convection pattern. While for a small layer thickness of 100 km, we observe a two-cell convection pattern with two upwellings stable in time that carry some amount of water through the layer, for the largest thickness (H=400 km), no upwelling water is observed. Moreover, the average temperature at the bottom boundary is below the melting point (cf. the grey circle in the bottom left panel of Fig. 4) and no water is present at the bottom boundary (cf. the grey circle in the bottom right panel) for this large thickness. This effect of the HP ice layer thickness H is similar to that of the reference viscosity μ_0 in a sense that it significantly changes the Rayleigh number - indeed, we can verify from Table 3 that Rayleigh numbers of simulations with a small thickness (# 7) and large melting point viscosity (# 3) are quite similar (10^9 and 1.6×10^9, respectively), as are those of large thickness (# 15, Ra=2.5×10^{11}) and small melting point viscosity (# 2, Ra=1.6×10^{11}).

3.4. Heat flux from the silicates

We have considered four values of heat flux from the silicate mantle: 10, 20, 30 and 40 mW m^{-2} together with different values of HP ice layer thickness. Note that even though in reality these values are not independent, here we treat them as such in order to investigate the effect of each parameter alone. Our results indicate that the heat flux from silicates does not influence the convection pattern since it does not significantly change the Rayleigh number. On the other hand, it has an important effect on the average bottom temperature and the amount of melt produced at the bottom boundary. Fig. 5 depicts two pairs of steady-state average temperature profiles. In the case of reference thickness (H=200 km, left panel), decreasing the heat flux from silicates to 10 mW m^{-2} (blue) results in the average bottom boundary temperature being below the melting point ($T^s_{av} < T^s_m$), contrary to the reference value of 20 mW m^{-2} (red), where a temperate layer is present at the interface with the silicates ($T^s_{av} = T^s_m$). Analogously, for the largest thickness (H=400 km, right panel), increasing the heat flux from silicates to 40 mW m^{-2} (cyan) results in the average bottom boundary temperature reaching the melting point ($T^s_{av} = T^s_m$), contrary to the reference value of 20 mW m^{-2} (red). The presence of a temperate layer at the silicate/HP ice interface is important because the liquid may interact with the silicates and leach volatiles and salts.

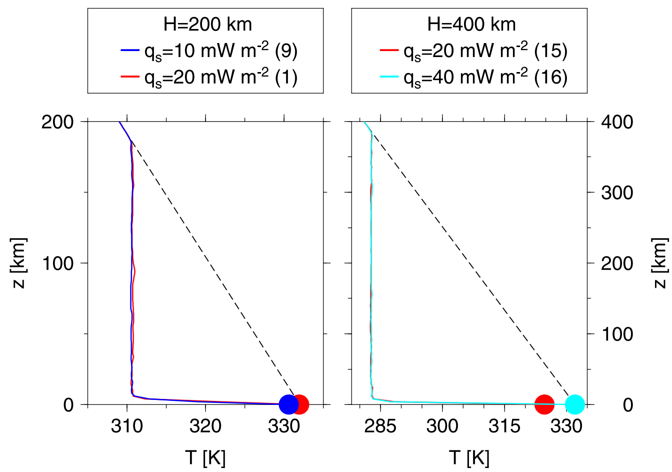

Fig. 5. Steady-state average temperature profiles for H=200 km, q_s=10 and 20 mW m^{-2} (left) and H=400 km, q_s=20 and 40 mW m^{-2} (right). Numbers in parentheses refer to simulation numbers from Table 3. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article).

3.5. Percolation threshold

We now investigate the effect of percolation threshold that controls the amount of meltwater extracted into the ocean. We have considered three distinct values: ϕ_c=1% (reference value), 3%, and 5%, which is the value observed for terrestrial sea ice (simulations # 1 and 17–18 in Table 3). Fig. 6 shows the temperature and porosity and the corresponding profiles in a statistical steady-state. We can see that with the increasing porosity threshold (top to bottom), the permeability of the top boundary decreases significantly – while for the smallest percolation threshold value of 1% the majority of water is extracted into the overlying ocean and the temperate layer is only ∼16 km thin (cf. Table 3), for larger values (3–5%), a few tens of kilometers thick temperate layer with a few percents of porosity is established. The value of percolation threshold thus controls the temperate layer thickness. Closer investigation of porosity profiles reveals that the average porosity at the top boundary is strictly controlled by the threshold value. The increase in percolation threshold also leads to the HP ice layer warming because less water is extracted through the top boundary.

3.6. Initial temperature

Finally, we have investigated the effect of the initial temperature condition on our results. In the reference setting, the initial temperature equals the melting temperature at the top interface with the ocean, T_0=T_m^o. We have added two simulations where the initial temperature equals the pressure-dependent melting temperature, T_0=$T_m(P)$ for H=200 and 400 km (simulations # 19–20 in Table 3). Even though initially the HP ice layer is much warmer in the two new simulations, the layer quickly cools down and reaches the steady-state average temperature (same for both initial conditions) within ∼2 Myr for H=200 km and ∼7 Myr for H=400 km. Fast cooling of the layer is caused by massive melting within an upwelling plume that appears at the beginning of the simulation and by the subsequent water extraction into the ocean.

3.7. Results synthesis

Our results demonstrate the importance of melting processes on the HP ice layer dynamics. Fig. 7 shows the total power consumed on melting/freezing in the whole domain (R_S=$\int_S r\,L\,\mathrm{d}S$) at a statistical steady-state as a function of power that is coming from the silicate mantle (Q_s=$2\,H\,q_s$). The very good correlation of these two quantities indicates that in steady-state all the heat supplied from silicates is used to melt the ice in the HP ice layer.

Fig. 8 depicts the time evolution of the average temperature at the silicate/HP ice interface for most of the simulations from this study. It shows that the steady-state average temperature at the bottom boundary is below the melting temperature (T_{av}^s<T_m^s) only for small viscosity (# 2), small heat flux (# 9), or large thickness (# 14 and 15). In all other simulations, the average temperature at the bottom boundary equals the melting temperature (T_{av}^s=T_m^s). Simulation # 12 lies in between these two cases. It is worth noting that the maximum temperature at the bottom boundary always equals the melting temperature and thus some amount of melt may be produced in all cases. However, if the whole boundary is not at the melting temperature (T_{av}^s<T_m^s), melt only appears in small transient patches, while if the whole boundary reaches the melting temperature, a stable and spatially continuous layer of ϕ_{av}∼1% is created at the silicate/HP ice interface.

Based on the results described above and in the previous sections, we distinguish three different melt patterns within the HP ice layer:

1. *Direct connection between the silicates and the ocean*
The average temperature at the silicate/HP ice interface equals the melting temperature (T_{av}^s=T_m^s) resulting in a formation of stable and continuous temperate layer. The generated melt is then advected through the convecting interior by the upwelling plumes that are stable in time and space and their temperature matches the melting curve (cf. the red curves in the bottom left panel of Fig. 3 and top left panel of Fig. 4). More melt might be generated during the ascent and also within the temperate layer at the ocean/HP ice interface. Eventually, the majority of melt is extracted into the ocean. This pattern that ensures a direct connection between the silicates and the ocean is characteristic for a small value of Rayleigh number (i.e. small HP ice layer thickness or large melting point viscosity).
2. *Indirect connection between the silicates and the ocean*
In the reference case, the average bottom boundary temperature equals the melting temperature (T_{av}^s=T_m^s) and a continuous temperate layer is formed at the silicate/HP ice interface. However most of the melt generated at this interface freezes during the ascent through the relatively cold convecting interior (cf. Fig. 2). Then melting occurs again in the temperate layer at the interface with the ocean. In this intermediate case, we predict an indirect connection between the silicates and the ocean.
3. *No connection between the silicates and the ocean*
For large values of Rayleigh number (i.e. large HP ice layer thickness or small melting point viscosity), the heat transfer by solid-state thermal convection is so efficient that the average temperature at the silicate/HP ice interface is below the melting point (T_{av}^s<T_m^s) and no stable temperate layer is formed. Isolated patches of melt may appear transiently but they usually freeze quickly. Similarly, no stable temperate layer at the bottom boundary was observed for most of the simulations with the smallest value of heat flux from silicates considered in this study (q_s=10 mW m^{-2}). For these simulations, we predict no connection between the silicates and the ocean as the occasional appearance of melt at the bottom boundary is not likely to result in significant salts/volatiles leaching.

In Fig. 9, we plot our results in terms of the identified melt pattern as a function of Rayleigh number and the heat flux from silicates. We observe that the connectivity between the silicates and the ocean decreases with the increasing vigor of convection (increasing Rayleigh number). However, the value of heat flux from silicates moves the pattern laterally resulting in no connectivity

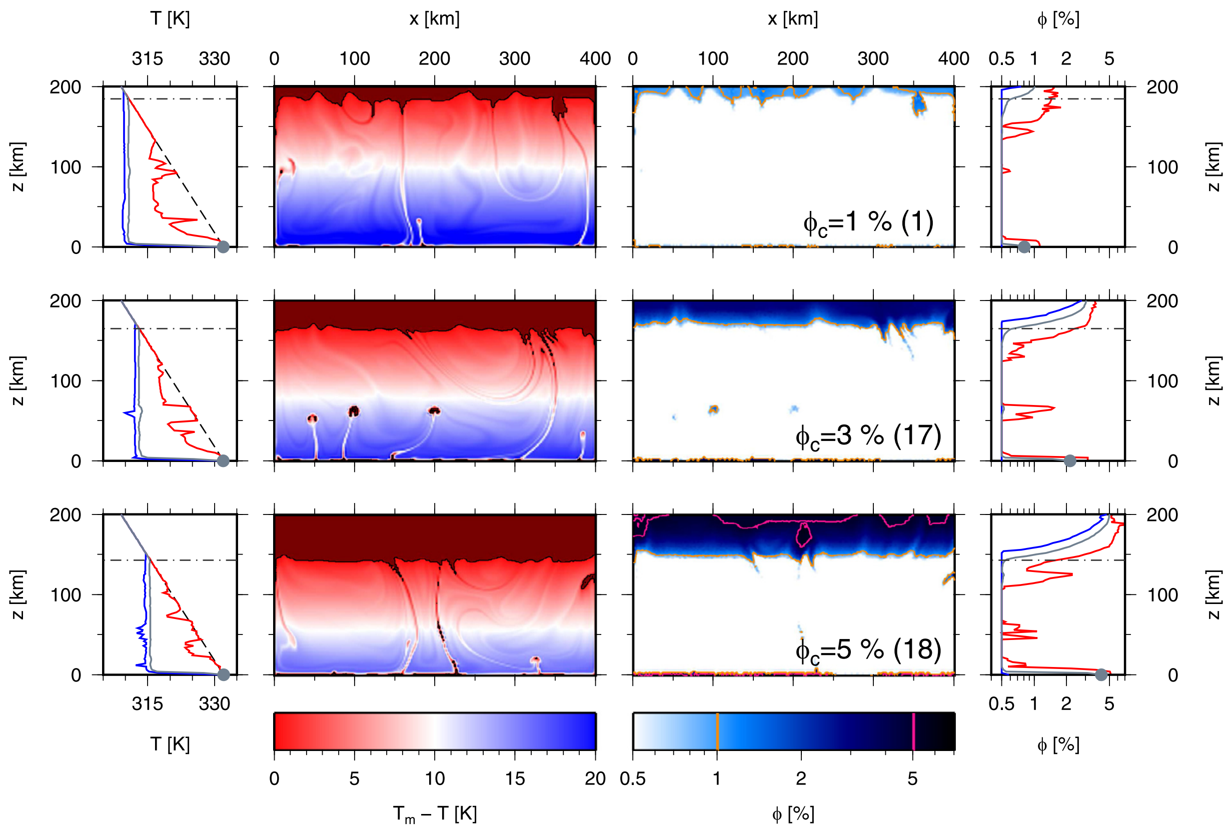

Fig. 6. The same as in Fig. 3 but for different values of percolation threshold (1–5%, top to bottom, numbers in parentheses refer to simulation numbers from Table 3). Pink contour in porosity panels marks ϕ=5%. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

regime (3) for intermediate value of Ra (# 9) and indirect connectivity regime (2) for large values of Ra (# 13 and 16).

As the satellite cools down, the HP ice layer thickens in time (e.g. Kirk and Stevenson, 1987; Bland et al., 2009) which corresponds to moving from left to right in Fig. 9 for a fixed viscosity at the melting point. In addition, the decay of radioactive elements leads to smaller values of the heat flux at the silicate/HP ice interface, which gives a general trend with less melt produced at the interface. Our results thus suggest that the connectivity between the silicates and the ocean decreases in time, i.e. the melt pattern potentially changes from direct connectivity (1) to indirect connectivity (2) to no connectivity (3).

Viscosity has the opposite effect: larger viscosity favors partial melt at the silicate/HP ice interface and a direct connectivity

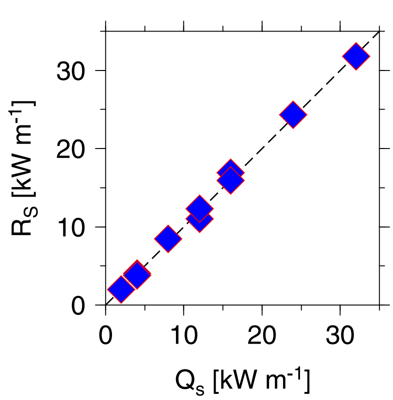

Fig. 7. Comparison of power consumed on melting/freezing, $R_S = \int_S r\,L\,\mathrm{d}S$ (computed at the statistical steady-state) with the power coming from silicates, $Q_s = 2\,H\,q_s$. For the reference value of Q_s (8 kW m^{-1}, relevant for simulations # 1–5, 8, 14, and 17–19) we plot the average value.

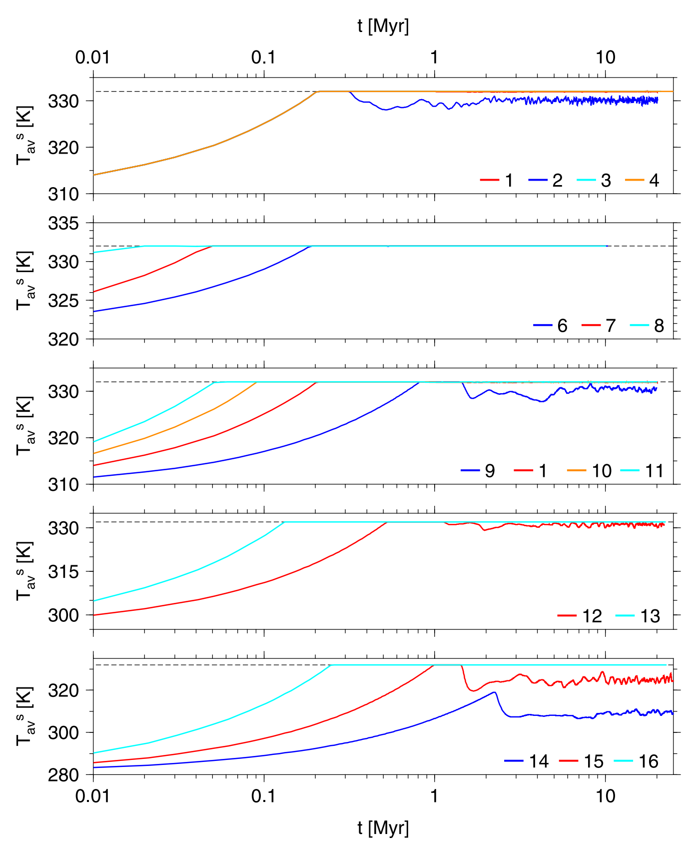

Fig. 8. The average temperature at the silicate/HP ice interface as a function of time for different simulations - the numbers refer to Table 3.

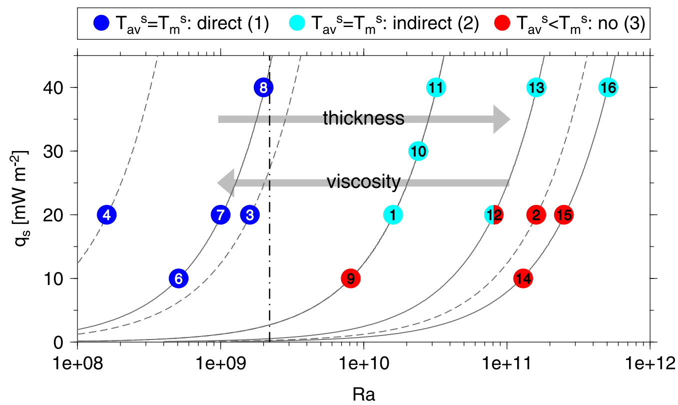

Fig. 9. The melt pattern identified in our results as a function of Rayleigh number and the heat flux from the silicate mantle: each dot represents one of our numerical simulations, the colors indicate the observed melt pattern (cf. text for definition), the numbers correspond to simulation numbers from Table 3. The full lines depict the (Ra, q_s) curves for fixed μ_0=10^{15} Pa s and the four thicknesses considered in this study (H=100–400 km); the dashed lines depict these curves for fixed H=200 km and the considered melting point viscosities (μ_0=10^{14}–10^{17} Pa s). The arrows indicate the effect of increasing layer thickness – less connectivity – and increasing viscosity – more connectivity. The vertical dot-dashed line marks the highest value of Ra investigated in Choblet et al. (2017). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

regime (1). The present study investigates a large range of viscosity values with the largest one being an extrapolation of laboratory measurements performed at very high shear stress of 1 MPa and higher to convective stresses several orders of magnitude smaller (cf. Appendix A). At very high values of shear stress, the strain rate and viscosity depend strongly on shear stress with the stress exponent larger than 4 for ice VI (Durham et al., 1996). However, all materials, including ice I and silicates, show much smaller value of the stress exponent when the material deforms under small shear stresses. Therefore, values of 10^{14} Pa s (similar to the ice I behavior, cf. Appendix A) to 10^{15} Pa s are more likely to describe the viscosity of ice VI at the melting point under convective stresses. If viscosity is 10^{14} Pa s, then only a very thin layer of ice VI (~ 100 km) would enable substantial melting at the interface with the silicate mantle (Ra ~ 10^{10}, regime 2). For a value of viscosity at the melting point equal to 10^{15} Pa s, exchange processes might be possible for thicknesses of up to 300 km (Fig. 9).

According to structural and thermal evolution models of Ganymede (Bland et al., 2009; Vance et al., 2014), the thickness of Ganymede's HP ice layer is expected to exceed 400 km at present. Unless the ice viscosity is abnormally high (> 10^{16} Pa s), it seems unlikely that significant melting occurs in the deep interior of Ganymede at present. However, as we demonstrated above, melting processes may play a key role for thinner ice layers. Thermal evolution modeling that would include such melting processes is now needed to address for how long the conditions favorable for melt generation and transport have been met during Ganymede's history.

4. Discussion

4.1. Comparison with previous results

Only a very few studies have attempted to address the problem of convection in the HP ice layers (Bercovici et al., 1986; Sotin and Parmentier, 1989; Coradini et al., 1995), and to our knowledge, only in the recent study of Choblet et al. (2017) the consequences of the HP ice layer dynamics for internal melting have been investigated. In that paper, a three-dimensional spherical geometry was considered, however, an immediate melt extraction was assumed which resulted in a practical condition that the amount of melt did not need to be tracked. It is therefore complementary to the present simulations that were performed only in a two-dimensional Cartesian geometry but while considering both phases, ice matrix and liquid water. Let us also note that Rayleigh numbers investigated in the present study (1.6×10^8–5.1×10^{11}, cf. Table 3 and Fig. 9) cover a larger range than what was considered in Choblet et al. (2017) (1.8×10^7–2.2×10^9, cf. also the vertical dot-dashed line in Fig. 9).

In general, the melting of ice in the HP ice layer of a Ganymede-like body is supported by both studies. Our results indicate that more melt is generated at the silicate/HP ice interface for small values of Rayleigh number while for large Rayleigh numbers the average temperature at this interface is below the melting temperature resulting in no basal melting. This finding is in agreement with the results displayed in Fig. 4c of Choblet et al. (2017) which shows that the basal melting rate decreases with increasing layer thickness (increasing Ra). Overall, our results suggest less bottom melting than what was found in Choblet et al. (2017) which is caused by higher values of Rayleigh number investigated in the present study. Finally, the values of Rayleigh numbers investigated in Choblet et al. (2017) fall in the direct connectivity regime found in the present study (cf. Fig. 9) which supports the validity of the instantaneous melt extraction assumption used in that study.

The effect of melting on solid-state thermal convection has already been investigated in the context of terrestrial planets (Ogawa et al., 2011; Ogawa, 2014; Ogawa and Yanagisawa, 2014). Although the setting may seem similar, there are several differences between those papers and our study. In Ogawa's work, internal heating is predominant which practically eliminates the hot thermal boundary layer and results in passive global upwelling with no active hot plumes (cf. also Parmentier et al., 1994). The decompression-melting leads to magma generation which then focuses the passive upwelling into active plumes due to the positive buoyancy of the bulk material caused by the presence of low density liquid within the solid matrix (magmatism and mantle upwelling (MMU) feedback, cf. Ogawa, 2014). The MMU feedback mechanism thus has the capacity to form active upwelling plumes in a convection pattern that would be devoid of such hot instabilities if it were not for the presence of buoyant partial melts. However, in our work, no internal heating is considered in the HP ice layer of Ganymede, which is only heated from below. Consequently, a hot thermal boundary layer, from which active hot upwelling plumes originate, is developed at the interface with silicates in all our simulations, and these hot instabilities would form even in the absence of melting. While the presence of partial melt in the plume head indeed enhances buoyancy and accelerates the upwelling, this effect is less dramatic than the MMU feedback mechanism described in Ogawa (2014).

4.2. Model limitations

In the energy balance (Eq. 1e), we neglect the advection of heat by liquid water ($\rho_w c_p^w \phi \boldsymbol{v}_w \cdot \nabla T$) by considering the zero porosity limit of a full two-phase mixture energy balance. This was motivated by the fact that majority of the HP ice layer is below the melting point where $\phi \sim 0\%$. As the porosity in our simulations is small (typically not exceeding a few percents) and the water velocity only deviates from the ice velocity at the top boundary during water extraction into the ocean, we believe that the zero porosity limit gives satisfactory results. To verify, we have performed a test by running the reference simulation (# 1 in Table 3) while using the energy balance with the full advection term. We found that the steady-state results (not shown here) agree very well with the reference solution and we thus confirmed that using Eq. (1e) does not change the solution significantly.

We also do not consider adiabatic decompression. The adiabatic gradient is $(\frac{\partial T}{\partial r})_S = -\frac{\alpha T g}{c_p} \sim -0.026\,\mathrm{K\,km^{-1}}$, about four times smaller than the temperature gradient across the layer: $(\frac{\partial T}{\partial r}) = -0.115\,\mathrm{K\,km^{-1}}$ (for the reference case). We have also performed an additional test with the adiabatic decompression effects taken into account. Our results (not shown here) indicated that the inclusion of adiabatic decompression leads to a slightly warmer interior since (i) less melting occurs in the upwelling plumes (because of cooling due to decompression) and (ii) the bottom part is warmed up by the compression of descending material. Less melting in the upwelling plumes also leads to a later establishment of the top temperate layer, however, in steady-state, the thickness of this layer is still controlled by the value of percolation threshold (cf. Fig. 6). Even though the potential for melting at the silicate/HP ice interface is slightly enhanced by the warmer interior, the addition of adiabatic decompression does not change the main conclusions of our study.

In our model, we also assume that the density difference between the two phases ($\Delta\rho$) is constant throughout the HP ice layer (and taken equal to the average between the bottom and top interface values, cf. Table 2). Evaluating the density differences between the two phases on both interfaces, we find that the value at the silicate/HP ice interface, $\Delta\rho^s = \rho_i^s - \rho_w^s$, is smaller than the average that is used in the simulations and, similarly, the value at the ocean/HP ice interface, $\Delta\rho^o = \rho_i^o - \rho_w^o$, is larger than the average. In other words, we overestimate the fluid buoyancy at the bottom boundary with silicates and underestimate it at the interface with the ocean. For thinner shells ($H \leq 200$ km), this error in density difference is less than 15% at both interfaces. For thicker shells ($H \geq 300$ km), this error exceeds 20% but only at the bottom interface with the silicates. Since our results indicate that melting is less likely at the silicate/HP ice interface of these thick layers, we believe that the assumption of $\Delta\rho$=const. does not significantly affect our conclusions.

In this study, we only consider ice VI even though the transition to ice V might be relevant. In such a case, the HP ice layer would be very thick (more than 400 km) and, according to the present results, there would likely be no melting at the bottom boundary. Since this transition is mostly pressure dependent, we do not expect that it would change our main results substantially. However, the effect of this phase transition will probably be more relevant for Titan because the pressure at the silicate/HP ice interface is about half the value for Ganymede and is thus much closer to the triple point ice VI–water–ice V.

In all our simulations, we considered pure water ice. However, leaching of salts/ammonia from the silicates, whether by water molten from the HP ice or during differentiation, would affect the HP ice layer dynamics in two ways: (i) the low-eutectic constituents could lower the melting point by several degrees (e.g. Journaux et al., 2013) which would lead to an earlier onset of melting, and (ii) the presence of salts would also change the ice and water densities and thus their relative buoyancy (Journaux et al., 2013; 2017). While being beyond the scope of this work, the role of salts/ammonia should be considered in future study.

4.3. Resolution tests and aspect ratio

In order to test the sufficiency of our computational mesh, we performed two simulations with H=200 and 300 km on a mesh with 400×200 elements and a corresponding refinement on the bottom boundary. This finer mesh has 366,000 cells and the bottom boundary layer resolution is 0.5 km for H=200 km and 0.75 km for H=300 km. The results of these simulations (not shown here) indicate that the steady-state convective picture does not depend on the chosen resolution, even though steady-state may be reached earlier for better resolution. The thermal profiles computed for both meshes agree reasonably well in both cases (H=200 and 300 km). The amount of melt at the bottom boundary is slightly resolution-dependent, with finer mesh leading to more melt. To reduce the effect of mesh resolution on our results, we have defined the particular melting patterns only in terms of temperature which is resolution independent.

To test the effect of the aspect ratio, we have performed one additional simulation with $\mu_0 = 10^{16}$ Pa s while keeping the other parameters on a reference value (corresponding to simulation # 3 from Table 3) in the box with aspect ratio 4 on a mesh consisting of 400×100 elements in the global grid and the same bottom refinement as in all other simulations. We have chosen this particular set of parameters because the corresponding Rayleigh number is small and the convective picture is stable (cf. the bottom row of Fig. 3). The results of this additional simulation (not shown here) suggest that the aspect ratio has no effect on the melting pattern defined in Section 3.7: the generation of melt at the bottom boundary and a stable upwelling that caries the meltwater upwards is observed in both cases (with aspect ratio 2 and 4). Also, the average temperature profiles match very well each other.

4.4. Tidal dissipation

Owing to the very low value of Ganymede's orbital eccentricity and its large distance from Jupiter, tidal dissipation in the HP ice layer is expected to be negligible at present. However, it is possible that Ganymede experienced prolonged periods of high eccentricity in the past. As proposed by Malhotra et al. (1991) and further explored by Showman and Malhotra (1997), Showman et al. (1997) and Bland et al. (2009), the Galilean satellites may have passed through one or more Laplace-like resonances before evolving into the current Laplace resonance. Passage through such a resonance may have excited Ganymede's eccentricity to values ranging between 0.01 and 0.015 during billions of years.

Assuming Maxwell viscoelastic rheology, ice viscosity of 10^{14} Pa s (minimal value considered in the present study), and the above eccentricity values, the dissipation rate in the HP ice layer would fall between 0.5 and $5 \times 10^{-8}\,\mathrm{W\,m^{-3}}$ (Tobie et al., 2005). For HP ice layer thicknesses ranging between 100 and 400 km, this would correspond to maximum integrated tidal heat flux ranging between 0.5 and 20 $\mathrm{mW\,m^{-2}}$. Even if tidal dissipation is concentrated in hot thermal upwellings due to the temperature-dependent viscosity (e.g. Tobie et al., 2003; Běhounková et al., 2012), the consequences for melting are expected to be rather small as long as the additional tidal heat flow does not exceed a few $\mathrm{mW\,m^{-2}}$, which remains small compared to the advective heat flow. Tidal heating may become important only for the upper value of the above estimates (10–20 $\mathrm{mW\,m^{-2}}$). In this particular case, coupled models that include temperature-dependent tidal heating are required to address the consequences on melt generation. We should, however, keep in mind that such high values of dissipation required an eccentricity ten times larger than the present-day value, and the duration of periods with elevated eccentricity are still uncertain. Tidal dissipation may also occur in the silicate mantle but again, the contribution to the heat budget would remain small and would not significantly affect the heat flux coming out of the silicates.

4.5. Comparison with Titan

Although Titan and Ganymede have very similar masses and radii, Titan has larger moment of inertia (Iess et al., 2012). Interior structure models consistent with this value indicate that the radius of Titan's silicate core is about 250 km larger than Ganymede's (Castillo-Rogez and Lunine, 2010). Moreover, Titan's radius is about

50 km smaller than Ganymede's radius, which results in the thickness of the H_2O layer being possibly less than 500 km on Titan (Lefèvre et al., 2014) and more than 800 km on Ganymede. The use of the same approach as for Ganymede to calculate the radii of Titan's internal interfaces results in much thinner HP ice layer (Sotin and Kalousová, 2016). The consequences of a smaller HP ice layer thickness (while assuming similar values of HP ice viscosity) are that more melt is produced at the silicate/HP ice interface and that volatiles and salts present at this interface are more likely to be transported to the ocean. To first order, such a thin permeable HP ice layer may explain the presence of ^{40}Ar, which comes from the decay of ^{40}K probably present in the silicate core, and methane in Titan's atmosphere. However, a Titan dedicated study is required in order to assess the potential for the HP ice layer to transfer volatiles and salts from the silicate/HP ice interface to the ocean and above into the atmosphere.

5. Conclusions and perspectives

In the present study, we investigated the dynamics of the deep high-pressure ice layer of Ganymede. In order to address the role of liquid water that is expected to be present within this layer we adopted the two-phase formalism, which is capable of handling the melting processes as well as the subsequent melt transport. We performed a series of two-dimensional simulations in Cartesian geometry, focused on (i) addressing the question of melt generation at the silicate/HP ice interface and (ii) investigating the evolution of connectivity between this interface and the ocean/HP ice interface.

Our results demonstrate that melt can be generated at the silicate/HP ice interface for small HP ice layer thickness and high values of heat flux and/or viscosity (small value of Rayleigh number). If generated, the meltwater is transported through the HP ice layer by the upwelling hot plumes and it may freeze and melt during the ascent. Melting also occurs within the temperate layer below the interface with the ocean from where it is extracted by porous flow into the ocean. The amount of extracted water as well as the temperate layer thickness is controlled by the assumed permeability model (value of percolation threshold).

Depending on the model parameters, three different melt patterns were identified: (1) direct connection of the silicates and ocean ensured by the liquid water ascending through a stable hot upwelling (small Rayleigh numbers), (2) indirect connection where the liquid water freezes and melts again during the ascent (reference values of model parameters), and (3) no connection between the silicates and the ocean (large Rayleigh numbers). Since the satellite's global cooling results in the thickening of the HP ice layer and the decrease of the heat flux from the silicates, our results suggest that the exchange of water (and salts/volatiles possibly leached from the silicate mantle) was more probable during the early stages of Ganymede's evolution and that the HP ice layer might have become less permeable as the moon cooled down.

Further studies should focus on the global thermo-chemical evolution of Ganymede to help address the exchange processes between its deep interior and surface. These results will be relevant for future exploration missions such as JUICE (Grasset et al., 2013). Application of our model on the HP ice layer of Titan which is probably thinner than Ganymede's might help explain the presence of ^{40}Ar in Titan's atmosphere as well as the striking difference between these two moons.

Acknowledgments

The authors would like to thank Francis Nimmo, Masaki Ogawa, and an anonymous reviewer for their valuable comments which helped to improve the quality of the manuscript. Part of this work was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration. KK acknowledges support by the office of the JPL Chief Scientist. This research received funding from the Czech Science Foundation through project 15-14263Y (KK) and was also supported by The Ministry of Education, Youth and Sports from the Large Infrastructures for Research, Experimental Development and Innovations project 'IT4Innovations National Supercomputing Center - LM2015070'. This work was partially supported by the Icy Worlds node of NASA's Astrobiology Institute (13-NAI7-0024). The research leading to these results has received financial support from the European Research Council under the European Community's Seventh Framework Programme FP7/2007–2013, Grant Agreement no. 259285 (GC, GT, OG). GC, GT and OG also benefited from CNES funding in the preparation of the JUICE mission.

Appendix A. Viscosity of ice VI

The viscosity of ice VI has been measured by two groups. Sotin et al. (1985) used a sapphire anvil cell to observe the ice VI deformation at large values of shear stress (typically between 50 and 100 MPa). They investigated the creep behavior at temperatures close to the melting temperature (253–293 K) and at pressures between 1.0 and 1.67 GPa, which is close to the transition to ice VII. In this limited range, the stress-dependence is small with the value of stress exponent equal to 1.9. Another group (Durham et al., 1996; Durham and Stern, 2001) measured the viscosity of ice VI in a gas-medium high-pressure deformation rig. The confining pressure was lower than that for Sotin et al.'s experiments, typically between 0.63 and 0.8 GPa, which is close to the transition to ice V, and the temperatures were also lower, between 200 and 250 K. The shear stress was typically between 1 and 5 MPa and the deformation rate was found to be much more stress-dependent with a stress exponent equal to 4.5.

At $\sigma = 1 \times 10^8$ Pa, the extrapolation of laboratory experiments performed by Sotin et al. (1985) provides a deformation rate of 10^{-2} s^{-1} at the melting point (filled black circle in Fig. A.10). Extrapolation of the laboratory experiments of Durham et al. (1996) along the melting curve provides deformation rates of 10^{-4} to 10^{-7} s^{-1} (diamonds in Fig. A.10). The deformation curve determined by Sotin et al. (1985) provides extrapolated values of deformation rate at the lower shear stresses investigated by Durham et al. (1996) which are very close (compare the short-dashed black line with the diamonds in Fig. A.10). The major difference between the two sets of laboratory experiments is the value of the stress exponent. It is possible that the stress exponent found by Sotin et al. (1985) is lower because the experiments were carried out close to the melting point, conditions that may favor grain boundary sliding.

Extrapolation to natural conditions must be handled with care because natural shear stresses in convection processes are much lower, typically from 1 to 10 kPa, and the creep behavior may be different than that at high shear stresses. This effect is well documented for ice I: At large shear stress, the deformation is dominated by dislocation creep and the slope of the strain rate–shear stress curve is steep (long-dashed line in Fig. A.10). At about 1 MPa, the creep behavior may be controlled by grain-boundary sliding (Goldsby and Kohlstedt, 2001) or basal sliding accommodated by grain boundary migration (Montagnat and Duval, 2000; Duval and Montagnat, 2002). It therefore becomes grain-size sensitive and the stress exponent gets below 2 (Durham et al., 2001; Goldsby and Kohlstedt, 2001). The same change in the dominant process is likely to exist for ice VI but no laboratory experiments to date can confirm such behavior.

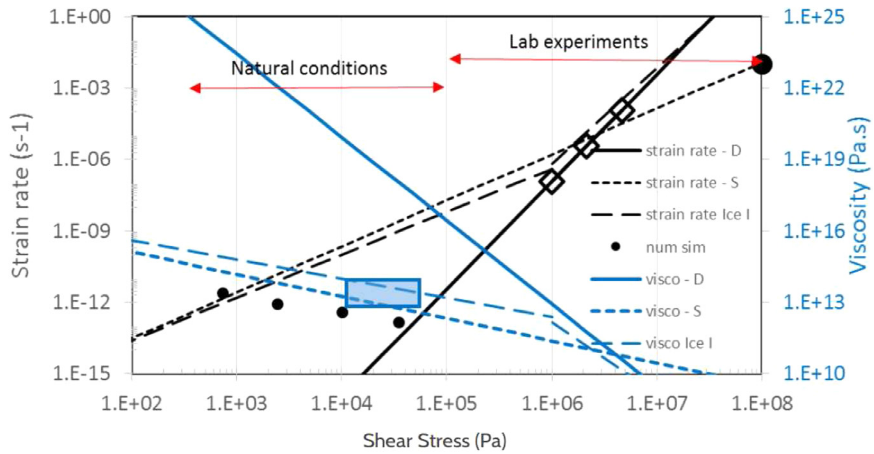

Fig. A.10. Strain rate (black) and viscosity (blue) of ice at its melting point as a function of shear stress for laboratory experiments, numerical simulations, and glacier measurements. The solid and short-dashed lines are for ice VI at P=1 GPa from extrapolation of laboratory measurements by Durham et al. (1996) and Sotin et al. (1985), respectively. The long-dashed line is for ice I with dislocation creep at high shear stress and grain boundary sliding at low shear stress. The filled black circle represents the (σ, $\dot{\varepsilon}$) of one experiment by Sotin et al. (1985) and empty diamonds are Durham et al.'s measurements. The blue rectangle shows the range of viscosity values measured in the terrestrial glaciers (Hudleston, 2015) and falls on top of the ice I viscosity curve. The small black dots correspond to rms values of (σ, $\dot{\varepsilon}$) from the numerical simulations # 1–4 in this study. The red arrows show typical range of shear stresses in convective layers (Natural conditions) and for the laboratory measurements. The large uncertainty in the value of HP ice viscosity leads us to investigate the dynamics of the HP ice layer for a wide range of viscosity values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Viscosity, the ratio of shear stress to strain rate, is represented in Fig. A.10 as blue lines. For ice I, the measurements on glaciers (blue box) where temperatures are close to the melting temperature cover the predictions from laboratory experiments on ice I (long-dashed blue line). Using the creep laws determined by the two groups cited above, one can calculate the value of viscosity along the melting curve. Both groups report that the isoviscous curves are almost parallel to the melting curve which justifies the use of Eq. (9). The present work investigates a large range of viscosity values due to our lack of knowledge on the creep behavior of ice VI at low stresses. The extrapolation from Sotin et al. (1985) provides values close to those of ice I (compare the short-dashed line and the long-dashed line in Fig. A.10). The extrapolation from Durham et al. (1996) suggests much larger values of viscosity (solid blue line) at similar values of the shear stress. Our numerical experiments (black points in Fig. A.10) for viscosities at the melting point between 10^{14} and 10^{17} Pa s provide values of the root mean square (rms) of the second invariant of shear stress and strain rate tensors that fall between the two experimental lines.

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